Lab Experiment - 2

Anshul(241210020)

Aim:

To design and verify the operation of an 8:3 Encoder and an 8:1 Multiplexer using basic logic gates and simulation tools such as Logisim.

1. 8:3 Encoder

An 8:3 Encoder is a combinational logic circuit that converts eight input lines into a 3-bit binary output. At any given time, only one input is assumed to be active.

Inputs:

- D0, D1, D2, D3, D4, D5, D6, D7

Outputs:

- Y2, Y1, Y0

Working Principle:

The encoder generates the binary equivalent of the active input line.

Boolean Expressions:

- Y2 = D4 + D5 + D6 + D7
- Y1 = D2 + D3 + D6 + D7
- Y0 = D1 + D3 + D5 + D7

For example:

- If D3 is HIGH, output will be 011
- If D6 is HIGH, output will be 110

The encoder reduces multiple input lines into fewer output bits, making it useful in data compression and digital communication systems.

2. 8:1 Multiplexer (MUX)

An 8:1 Multiplexer is a combinational circuit that selects one of eight input lines and forwards it to a single output line based on the selection inputs.

Inputs:

- I0, I1, I2, I3, I4, I5, I6, I7

Selection Lines:

- S2, S1, S0

Output:

- Y

Working Principle:

The selection lines determine which input is connected to the output.

Output Expression:

Y = I0S2'S1'S0' + I1S2'S1'S0 + I2S2'S1S0' + I3S2'S1S0 + I4S2S1'S0' + I5S2S1'S0 + I6S2S1S0' + I7S2S1S0

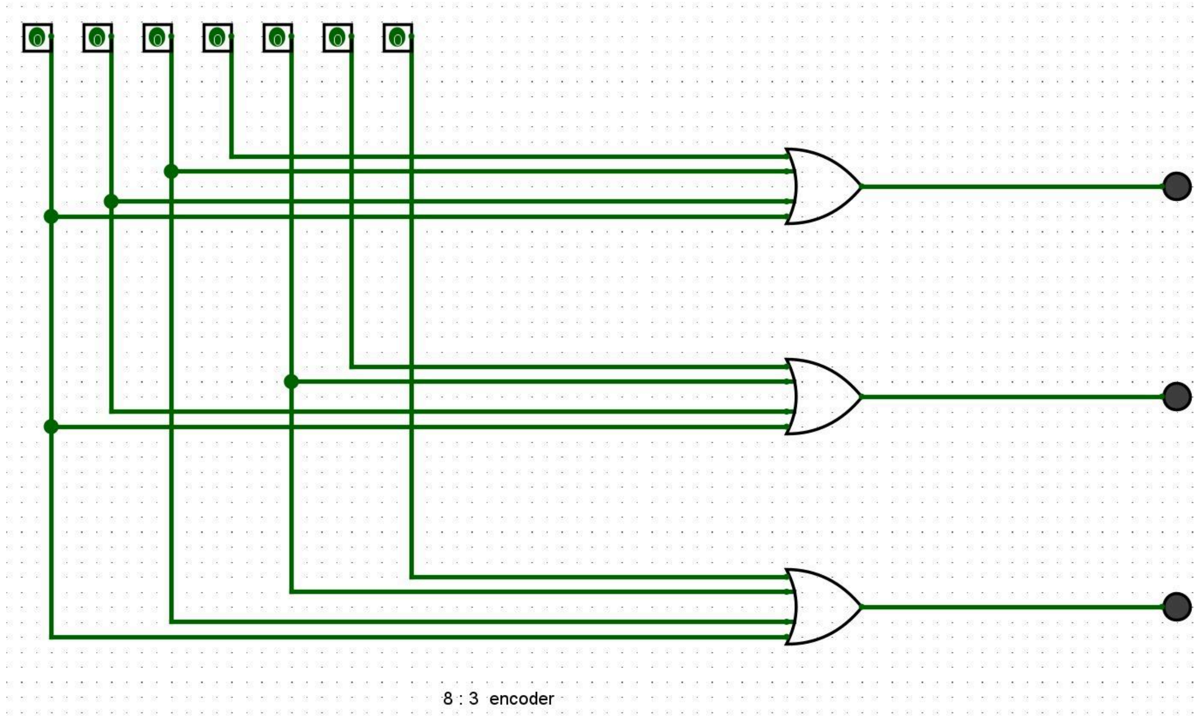
Depending on the binary value of selection lines:

- 000 selects I0
- 001 selects I1
- 010 selects I2
- ...
- 111 selects I7

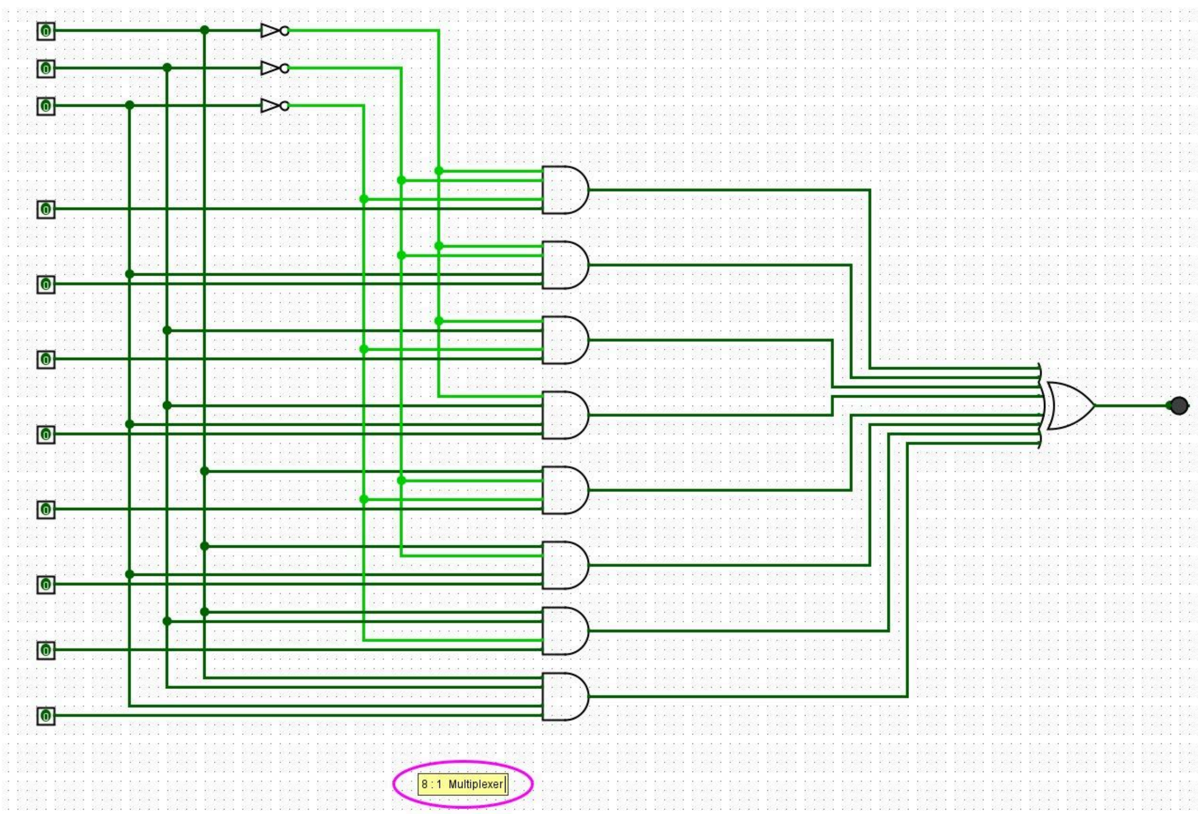
The Multiplexer acts as a digital data selector and is widely used in communication systems, arithmetic circuits, and control units.

Logisim Implementation:

Both the 8:3 Encoder and 8:1 Multiplexer circuits were implemented and tested in Logisim software to verify their truth tables and operational correctness.



8 : 3 encoder



8 : 1 Mutliplexer